AICR · SELF-CONTAINED NOTE

# *Capping the Fixed Set*

*The $P$-axis of the split-source decomposition, with the whole chain it rests on*

**Resolves.** *Decomposition for Split Unpredictable Sources* (`c/0010`). **Partially, and for a relaxed statement.** The conjecture's hypothesis on $P$ is removed entirely, on the region (H2), provided the mixture family may depend on the observer's query budget $q$. That is a relaxation of the conjecture, not the conjecture. The order-respecting statement, with an absolute constant and a $q$-free family, is *not* settled here.

**Status.** Self-contained modulo four published external results, which are isolated as Assumptions in §2 and are the only unproved inputs. Everything else is stated and proved below. Verification is *not* uniform across the chain, and §14 gives it per result: the final arm has had six independent blind referee passes, the arm it rests on has had none.

## Reader's guide

This note proves, on a region, a relaxation of a conjecture about presampling for two sources that read a random oracle independently. It is assembled from four campaign artifacts so that it can be read without them.

**The new content is §11, and it is short.** The conjecture asks for *some* family of $P$-mixtures; it never asks that the fixed sets be large, and Definition 2 bounds them by $|I| \leq P$ rather than fixing $|I|$. So when the requested $P$ exceeds what the proof can afford, do not use it — build the family at the balance point instead. A $P_0$-mixture with $P_0 \leq P$ *is* a $P$-mixture. That removes the hypothesis (H1) from Theorem 5 at a cost of $8 \to 13$ in the constant $C$.

**Everything before §11 is the chain.** §§3–7 build two extraction bounds and one lower bound; §8 constructs the family; §9 converts extraction into decomposition; §10 reduces the hypotheses. A reader who wants only the new result can go to §11 and take Theorem 5 on faith.

**Two things this does not prove, both overclaimed in an earlier draft before referees caught them.**

- The family must be allowed to depend on $q$, which Remark 2 forbids. So Theorem H proves a relaxation, and the window recorded as open — absolute $C$, $q$-free family — stays open.
- Keeping the family $q$-free costs a factor $\sqrt{q^+}$ in the constant (Theorem H′). It does not follow that the $\sqrt{q^+}$ *is* the price of $q$-independence: what is shown is that $q$-dependence *suffices* to remove it within this route. No lower bound against $q$-free families exists anywhere.

**What remains.** The hypothesis (H2), a condition on the output size $M$, is carried unchanged throughout. It is the whole residue on the $P$-axis; §12 is about what it means and whether applications satisfy it. They do for key extraction and do not for long-output extraction.

# 1 Setting

Fix $N, M \geq 2$ and let $\mathsf{Fun} := \{f : [N] \times [N] \to [M]\}$, with $H \leftarrow_\$ \mathsf{Fun}$. Logarithms are to base 2 and ln is natural.

**Definition 1 (sources, splitness, unpredictability).** A *source* is a computationally unbounded randomized algorithm $S$ with oracle access to $H$ and unbounded query count — so it may read all of $H$ — taking no input, using private coins, and outputting a pair $(x, z) \in [N] \times \{0,1\}^*$. Two sources $S_1, S_2$ are *split* if run on independent private coins, so that conditioned on $H$ their output pairs are independent. Write $(x_i, z_i) \leftarrow S_i^H$ with $|z_i| \leq \sigma_i$, and put $\mathbf{x} := (x_1, x_2)$, $\mathbf{z} := (z_1, z_2)$. A *predictor* $\mathsf{P}$ is computationally unbounded with oracle access to $H$ and unbounded query count, takes $\mathbf{z}$, and outputs an element of $[N]$. The pair is *$\delta$-unpredictable* if $\Pr\left[\mathsf{P}^H(\mathbf{z}) = x_i\right] \leq \delta$ for $i = 1,2$ and every $\mathsf{P}$. Since a predictor may ignore its input and output a fixed element, necessarily $\delta \geq 1/N$.

**Definition 2 (bit-fixing sources, and consistency).** A distribution $Y$ over $\mathsf{Fun}$ is *$P$-bit-fixing* if there are a set $I \subseteq [N] \times [N]$ with $|I| \leq P$ and a map $a : I \to [M]$ such that a draw $g \leftarrow Y$ satisfies $g(u) = a(u)$ for every $u \in I$ and is uniform on $[M]$ and independent across the points $u \notin I$. We call $I$ its *fixed set* and $a$ its *fixed values*, and say $Y$ is *consistent with* $f \in \mathsf{Fun}$ if $a(u) = f(u)$ for every $u \in I$. A *$P$-mixture* is a finite indexed family $(\lambda_j, Y_j)_{j \in \mathcal{J}}$ with $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$ and each $Y_j$ a $P$-bit-fixing source; it is *consistent with* $f$ if every component is. A *draw* means: draw $J$ with $\Pr\left[J = j\right] = \lambda_j$, then $g \leftarrow Y_J$.

The inequality $|I| \leq P$ is the hinge of §11 and is not a stylistic choice: Assumption 2's source states the same convention, "$P$-bit-fixing if it is fixed on *at most* $P$ coordinates".

**Definition 3 (observers, and the four experiments).** An *observer* is a computationally unbounded algorithm $D$ with oracle access, making at most $q$ queries, taking a pair in $[M] \times (\{0,1\}^*)^2$ and outputting a bit. It is *challenge-oblivious*: its input is $\mathbf{z}$ together with a single value of $[M]$, never $\mathbf{x}$. Given a family $\mathcal{Y} = \{Y_{f,\zeta}\}$ of $P$-mixtures indexed by $f \in \mathsf{Fun}$ and $\zeta \in (\{0,1\}^*)^2$:

- $\mathsf{Real}$: draw $H$; draw $(x_i, z_i) \leftarrow S_i^H$ on independent coins; set $y := H(\mathbf{x})$; output $D^H(y, \mathbf{z})$.
- $\mathsf{Real}_0$: as $\mathsf{Real}$ but $y \leftarrow_\$ [M]$.
- $\mathsf{Dec}$: draw $H$ and $(\mathbf{x}, \mathbf{z})$ as in $\mathsf{Real}$ and do not resample; draw $(J, H^*) \leftarrow Y_{H,\mathbf{z}}$; set $y^* := H^*(\mathbf{x})$; output $D^{H^*}(y^*, \mathbf{z})$. Write $I_J$ for the fixed set drawn.
- $\mathsf{Dec}_0$: as $\mathsf{Dec}$ but $y \leftarrow_\$ [M]$.

Put $\mathsf{Adv}_{\mathcal{Y},D} := |\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Dec} = 1]|$ and

$$\kappa(q) := \sup_D \big|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\big|,$$

the supremum over $q$-query challenge-oblivious $D$. Note $\kappa$ involves $\mathsf{Real}$ and $\mathsf{Real}_0$ only, so it does not depend on $\mathcal{Y}$: the oracle is the real $H$ in both and only the challenge changes. Write $\kappa^{\mathrm{na}}(q)$ for the same supremum restricted to observers of challenge resolution 1 (Definition 4).

Throughout,

$$\sigma := \sigma_1 + \sigma_2, \qquad \sigma' := \sigma + 2\log N, \qquad q^+ := q + 1, \qquad S := \sqrt{\sigma' q^+ \delta},$$

$$t_q := \sqrt{\sigma' q^+/\delta}, \qquad t_0 := \sqrt{\sigma'/\delta}, \qquad t_q = t_0\sqrt{q^+}, \qquad B := \sigma' + \log\gamma^{-1},$$

and $\mathcal{Z} := \{0,1\}^{\le\sigma_1} \times \{0,1\}^{\le\sigma_2}$, so $|\mathcal{Z}| < 2^{\sigma+2}$. Since $N \ge 2$ gives $2\log N \ge 2$ and $\sigma \ge 0$, we have $\sigma' \ge 2$ throughout; this is used repeatedly and is why $N \ge 2$ is a standing hypothesis.

**Conjecture 1 (the target).** *There are absolute constants $c, C$ such that for all $N, M$, every split $\delta$-unpredictable pair with leakage bounded by $\sigma_1, \sigma_2$, every $P \in \mathbb{N}$ and every $\gamma \in (0,1)$, there is a family $\mathcal{Y} = \{Y_{f,\zeta}\}$, $f \in \mathsf{Fun}$, $\zeta \in (\{0,1\}^*)^2$, depending only on $S_1, S_2, P, \gamma$, in which every $Y_{f,\zeta}$ is a $P$-mixture consistent with $f$, such that for every $q \in \mathbb{N} \cup \{0\}$ and every $q$-query challenge-oblivious $D$,*

$$\mathsf{Adv}_{\mathcal{Y},D} \le \frac{c(\sigma' + \log\gamma^{-1})q^+}{P} + C\sqrt{\sigma' q^+ \delta} + \gamma.$$

**Remark 1 (what the indexing allows).** The family is indexed by the oracle as well as the leakage — the one departure from the classical single-source lemma. It may *not* be chosen using $\mathbf{x}$, which is why the index set is $\mathsf{Fun} \times (\{0,1\}^*)^2$ and nothing larger.

**Remark 2 (quantifier order).** The order is part of the statement. The family is chosen after $P$ and $\gamma$ but *before* $q$, so no part of it may be tuned to the observer's query budget. Relaxing this makes the statement easier. Theorem H below takes exactly that relaxation and no other.

Two facts about $\mathbf{x}$ are used in §9. Both are the Contract's, and neither is reproved here.

**Lemma 1 (the fixed set misses the challenge).** *Let $\mathcal{Y}$ be any family as in Conjecture 1, let $J$ index the component drawn in* $\mathsf{Dec}$ *or* $\mathsf{Dec}_0$, *and let $I_J$ be its fixed set. Then $\Pr\big[\mathbf{x} \in I_J\big] \le P\delta$.*

**Lemma 2 (the observer misses the challenge).** *In* $\mathsf{Dec}_0$, *let $Q$ be the set of points $D$ queries. Then $\Pr\big[\mathbf{x} \in Q\big] \le q\delta$, the case $q = 0$ holding because $Q$ is empty.*

## 2 The external inputs

Four published results are used and not proved. They are isolated here so that the rest of the note is self-contained relative to them. Each is quoted from a source card rather than from the paper itself, which is a gap recorded in §14.

**Assumption ([CDGS, Claim 2], decomposition).** Let $\mu$ be a distribution on $\mathsf{Fun}$ with $S := N^2 \log M - H_\infty(\mu)$, let $\delta_0 \in (0,1]$ and $\gamma \in (0,1)$, and put $P := (S + \log\gamma^{-1})/(\delta_0 \log M)$. Then $\mu = \sum_j \lambda_j X_j + \lambda_{\mathrm{fin}} Y_{\mathrm{fin}}$ with $\lambda_{\mathrm{fin}} \le \gamma$, the supports pairwise disjoint, and each $X_j$ a $(P, 1-\delta_0)$-dense source fixing a set $I_j$ with $|I_j| \le P$ to values every $f$ in its support takes.

**Assumption ([CDGS, Claim 3], dense-versus-bit-fixing).** Let $X$ be a $(P, 1-\delta_0)$-dense source and $Y$ the $P$-bit-fixing source on the same fixed set and values. Every $q$-query adaptive distinguisher between $X$ and $Y$ has advantage at most $q\delta_0 \log M$.

**Assumption ([CFHS, Lemma 3], flat decomposition).** Let $k \in \mathbb{R}_{>0}$ with $2^k \in \mathbb{N}$. Every $k$-source $X$ — one with $\Pr[X = x] \le 2^{-k}$ for all $x$ — is a convex combination of flat $k$-sources, each uniform on a set of size exactly $2^k$.

**Assumption ([CFHS, Lemma 4(3)] and Hoeffding).** For $k, n \in \mathbb{N}$ with $0 < k \le n$, $\binom{n}{k} \le (en/k)^k$. And if $Y_1, \ldots, Y_n$ are independent, mean zero, each in an interval of length 1, then $\Pr[|\sum_i Y_i| \ge \lambda] \le 2\exp(-2\lambda^2/n)$.

The first is *item 3* of a three-item lemma titled "Numeric inequalities", on p. 9 of the ePrint. It is **not** numbered "Lemma 4.3", which exists in neither the ePrint nor the published version; a citation in that form is unresolvable. Corrected after checking the card against the paper.

## 3 Structural facts

**Lemma 3 (derandomisation).** *Fix one of the two paired experiments. For every $q$ and every $q$-query challenge-oblivious $D$ there is a deterministic such $D'$ with the relevant advantage no smaller for $D'$ than for $D$, and of the same challenge resolution. Consequently the suprema defining $\kappa(q)$ and $\kappa^{\mathrm{na}}(q)$ are attained on deterministic observers, the latter on deterministic observers of resolution* 1.

*Proof.* Let $\rho$ be $D$'s coins and $D_\rho$ the deterministic strategy it then runs. All of $\mathsf{Fun}, [N], [M], \mathcal{Z}$ are finite and $D$ makes at most $q$ queries, so there are finitely many deterministic $q$-query challenge-oblivious strategies. The quantity at issue is $|\mathbb{E}_\rho[\alpha(\rho)]|$ with $\alpha(\rho) = \Pr[\mathsf{E}_1 = 1 \mid \rho] - \Pr[\mathsf{E}_0 = 1 \mid \rho]$, the coins being independent of every other draw and the paired experiments differing in no other component; so it is at most $\max_\rho |\alpha(\rho)|$, attained at some $\rho^*$. Take $D' := D_{\rho^*}$. It inherits $D$'s resolution because Definition 4 quantifies over every coin string, so a property holding at every $\rho$ holds at $\rho^*$. □

Once $D$ is deterministic and its oracle fixed to $f$, its output is a function of its challenge input alone; write $\theta : [M] \to \{0,1\}$ for that function and $p_\theta := |\theta^{-1}(1)|/M$.

**Lemma 4 (posterior product structure).** *For every $(f,\zeta)$ in the support of $(H,\mathbf{z})$, the conditional law of $\mathbf{x}$ is the product $\Pi_{f,\zeta} = \pi^1_{f,\zeta_1} \otimes \pi^2_{f,\zeta_2}$ with $\pi^i_{f,\zeta_i}(u) = \Pr\left[x_i = u \mid H = f, z_i = \zeta_i\right]$. Writing $m_i := \|\pi^i_{f,\zeta_i}\|_\infty$: (i) $\max_{\mathbf{u}} \Pi_{f,\zeta}(\mathbf{u}) = m_1 m_2$; (ii) $\mathbb{E}[m_i] \leq \delta$; (iii) $\mathbb{E}[m_1 m_2] \leq \delta$; (iv) $\Pr[m_1 m_2 > \vartheta] \leq 2\delta/\sqrt{\vartheta}$ for $\vartheta \in (0,1]$.*

*Proof.* The sources run on independent private coins and on the same $H$, so conditioned on $H = f$ their output pairs are independent: $\Pr\left[x_1 = u_1, z_1 = \zeta_1, x_2 = u_2, z_2 = \zeta_2 \mid f\right] = \prod_i \Pr\left[x_i = u_i, z_i = \zeta_i \mid f\right]$. Summing over $u_1, u_2$ gives $\Pr\left[\mathbf{z} = \zeta \mid f\right] = \prod_i \Pr\left[z_i = \zeta_i \mid f\right] > 0$, and dividing gives the product form. (i) is that the maximum of a product measure is the product of the maxima.

For (ii), let $\mathsf{P}$ be the predictor that on input $\mathbf{z}$, with oracle access to $H$, reads all of $H$, computes $\pi^i_{H,z_i}$ — it may, the sources being fixed and predictors arbitrary — and outputs a maximiser. Then $\Pr\left[\mathsf{P}^H(\mathbf{z}) = x_i\right] = \mathbb{E}_{(f,\zeta)}[m_i(f,\zeta)]$, which unpredictability bounds by $\delta$. (iii) follows from $m_2 \leq 1$ and (ii). For (iv), $m_1 m_2 > \vartheta$ with both factors in $[0,1]$ forces $m_i > \sqrt{\vartheta}$ for some $i$; Markov and (ii) give $\Pr\left[m_i > \sqrt{\vartheta}\right] \leq \delta/\sqrt{\vartheta}$. □

Fact (i) is where the second source is spent: a single cell of $[N]^2$ carries mass $m_1 m_2$, a product of two independently small numbers, whereas one unpredictable source over $[N]^2$ gives only $\delta$ per cell.

**Remark 3 ($\mathbb{E}[m_1 m_2] \leq \delta^2$ is false).** This is needed later to justify an exponent. Let $2 \leq M \leq N$, $\delta := 1/M$, let $E$ be the event $f(1,1) = 1$, of probability $\delta$, and let both sources output $x_i := 1$ if $E$ holds and $x_i$ uniform on $[N]$ otherwise, leaking the empty string. The coins are independent, so the pair is split, and $\mathbb{E}[m_i] = \delta + (1-\delta)/N \leq 2\delta$, so the pair is $2\delta$-unpredictable; yet $m_1 m_2 = 1$ on $E$, so $\mathbb{E}[m_1 m_2] \geq \delta$. Splitness constrains the coins, not the two sources' common dependence on $f$.

**Lemma 5 (flattening).** *Let $\pi$ be a distribution on $[N]$ with $\|\pi\|_\infty = m$ and $k := \lfloor 1/m \rfloor$. Then $1 \leq k \leq N$, $1/k \leq 2m$, and $\pi$ is a convex combination of distributions uniform on subsets of size exactly $k$. Consequently, for every $G : [N]^2 \to [-1,1]$ and every $\pi^1, \pi^2$ with $\|\pi^i\|_\infty = m_i$, $k_i = \lfloor 1/m_i \rfloor$,*

$$\left|\mathbb{E}_{\pi^1 \otimes \pi^2}[G]\right| \leq \max\left\{ \left|\mathbb{E}_{\mathrm{Unif}(B_1) \otimes \mathrm{Unif}(B_2)}[G]\right| \ : \ |B_i| = k_i \right\}.$$

*Proof.* $m \geq 1/N$ gives $k \leq N$; $m \leq 1$ gives $k \geq 1$. For $x \geq 1$, $\lfloor x \rfloor \geq x/2$ — for $1 \leq x < 2$ because $\lfloor x \rfloor = 1 \geq x/2$, for $x \geq 2$ because $\lfloor x \rfloor > x - 1 \geq x/2$ — so with $x = 1/m$, $1/k \leq 2m$. If $k = 1$ the decomposition is the identity $\pi = \sum_u \pi(u)\delta_u$ and no external result is needed; assume $k \geq 2$. Since $\|\pi\|_\infty \leq 1/k = 2^{-\log k}$ with $\log k > 0$ and $2^{\log k} = k \in \mathbb{N}$, Assumption 2 applies. The displayed inequality follows because $(\rho^1, \rho^2) \mapsto \mathbb{E}_{\rho^1 \otimes \rho^2}[G]$ is bilinear, so substituting both decompositions writes the left side as a convex combination of the quantities on the right. □

Note $G$ is arbitrary; it will carry an indicator $\mathbb{1}[\mathbf{u} \notin \mathcal{S}]$ that does not factor over the coordinates, which is harmless because the linear structure used is in $(\pi^1, \pi^2)$, not in $G$.

**Lemma 6 (second-moment sharpening).** *With $m_1, m_2$ as in Lemma 4, and $t_c$ as in (1) below: (a)* $\mathbb{E}[\sqrt{m_1 m_2}] \le \delta$; *(b)* $\Pr[m_1 m_2 > \vartheta] \le \delta/\sqrt{\vartheta}$ *for* $\vartheta \in (0,1]$; *(c) pointwise, for* $k_i = \lfloor 1/m_i \rfloor$ *and every* $c > 0$, $t_c(k_1,k_2) \le \sqrt{L(m_1+m_2)} + \sqrt{2c}\sqrt{m_1 m_2}$; *(d)* $\mathbb{E}[t_c(k_1,k_2)] \le \sqrt{2L\delta} + \delta\sqrt{2c}$.

*Proof.* (a) $m_1, m_2$ are nonnegative and bounded by 1, so all expectations are finite, and by Cauchy–Schwarz and Lemma 4(ii) applied to each coordinate separately, $\mathbb{E}[\sqrt{m_1}\sqrt{m_2}] \le \sqrt{\mathbb{E}[m_1]}\sqrt{\mathbb{E}[m_2]} \le \delta$. (b) $m_1 m_2 > \vartheta$ iff $\sqrt{m_1 m_2} > \sqrt{\vartheta}$; Markov and (a). (c) $t_c^2 = \frac{\ln(eN/k_1)}{2k_2} + \frac{\ln(eN/k_2)}{2k_1} + \frac{c}{2k_1 k_2}$. Each $k_i \ge 1$, so $\ln(eN/k_i) \le L$; and $1/k_i \le 2m_i$, so $\frac{1}{2k_i} \le m_i$ and $\frac{1}{2k_1k_2} \le 2m_1m_2$. Hence $t_c^2 \le L(m_1+m_2) + 2c\, m_1 m_2$, and $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ gives the claim. (d) Take expectations in (c). By concavity of $\sqrt{\cdot}$ and Lemma 4(ii), $\mathbb{E}[\sqrt{L(m_1+m_2)}] \le \sqrt{L(\mathbb{E}m_1 + \mathbb{E}m_2)} \le \sqrt{2L\delta}$; and $\sqrt{2c}\,\mathbb{E}[\sqrt{m_1m_2}] \le \delta\sqrt{2c}$ by (a). □

Part (a) is what buys the sharpening. Bounding $\sqrt{L(\mathbb{E}m_1 + \mathbb{E}m_2) + 2c\,\mathbb{E}[m_1m_2]} \le \sqrt{2\delta(L+c)}$ by concavity on the whole expression, as the cruder route does, gives $\sqrt{2c\delta}$ where splitting first and applying Cauchy–Schwarz to the cross term gives $\delta\sqrt{2c}$. The stronger hypothesis $\mathbb{E}[m_1m_2] \le \delta^2$, which would give the same conclusion directly, is false by Remark 3, and (a) does not follow from $\mathbb{E}[m_1m_2] \le \delta$ by Jensen, which runs the other way.

# 4 Rectangle discrepancy

Put $L := \ln(eN)$ and, for $c > 0$,

$$t_c(k_1,k_2) := \sqrt{\frac{k_1 \ln\frac{eN}{k_1} + k_2 \ln\frac{eN}{k_2} + c}{2k_1k_2}}. \tag{1}$$

A *revealing rule of budget $s$* is a family $(\mathcal{A}_\zeta)_{\zeta\in\mathcal{Z}}$ of procedures, where $\mathcal{A}_\zeta$ inspects coordinates of $f$ one at a time — the next a function of $\zeta$ and the values already seen — halts having inspected a set $\mathcal{S}(f,\zeta)$ with $|\mathcal{S}(f,\zeta)| \le s$, and outputs a test $\theta_{\zeta,f} : [M] \to \{0,1\}$ that is a function of $\zeta$ and the inspected values.

The next two lemmas bound the same rectangle sum in two different ways, and the difference between them is the whole difference between the two extraction bounds of §6. Lemma 7 restricts the test class to the $|\mathcal{Z}| < 2^{\sigma+2}$ tests a revealing rule can reach, and pays for the revealed set. Lemma 8 quantifies over all $2^M$ tests, and pays $M$ instead — but owes nothing for revelation.

**Lemma 7 (restricted test class).** *Let $N \ge 2$, $\gamma_0 \in (0,1)$ and $(\mathcal{A}_\zeta)$ a revealing rule. Put $C_0 := (\sigma+2)\ln 2 + \ln\frac{4N^2}{\gamma_0}$. With probability at least $1-\gamma_0$ over $f \leftarrow_\$ \mathsf{Fun}$, simultaneously for all $\zeta \in \mathcal{Z}$, all $k_1,k_2 \in [N]$ and all $B_i \subseteq [N]$ with $|B_i| = k_i$, writing $R = B_1 \times B_2$,*

$$\left| \frac{1}{k_1k_2} \sum_{\mathbf{u}\in R\setminus\mathcal{S}(f,\zeta)} \left(\theta_{\zeta,f}(f(\mathbf{u})) - p_{\theta_{\zeta,f}}\right) \right| \le t_{C_0}(k_1,k_2).$$

*Moreover* $\mathbb{E}[t_{C_0}] \le \sqrt{2\delta(L+C_0)} \le \sqrt{8\delta(\sigma' + \log\gamma_0^{-1})}$.

*Proof.* **(1)** Fix $\zeta$. For $S_0 \subseteq [N]^2$ and $w \in [M]^{S_0}$ the event $\mathcal{T}_{S_0,w} = \{\mathcal{S}(f,\zeta) = S_0 \wedge f|_{S_0} = w\}$ is measurable with respect to $f|_{S_0}$: replaying $\mathcal{A}_\zeta$ on $w$ reproduces its whole inspection sequence. The coordinates of $f$ being i.i.d. uniform, conditioning on such an event leaves $f|_{[N]^2\setminus S_0}$ i.i.d. uniform; and $\theta_{\zeta,f}$, $p_{\theta_{\zeta,f}}$ are constant on $\mathcal{T}_{S_0,w}$.

**(2)** Fix also $k_1,k_2,B_1,B_2$ and condition on $\mathcal{T}_{S_0,w}$. Write $\theta, p_\theta$ for the constants and $n_0 := |R \setminus S_0| \le k_1k_2$. The variables $\theta(f(\mathbf{u})) - p_\theta$, $\mathbf{u} \in R \setminus S_0$, are independent, lie in an interval of length 1, and have mean 0, since $f(\mathbf{u})$ is uniform on $[M]$ and $\Pr\big[\theta(f(\mathbf{u})) = 1\big] = p_\theta$ exactly. Assumption 2 with $\lambda := k_1k_2t_{C_0}$ gives conditional failure probability at most $2\exp(-2\lambda^2/n_0) \le 2\exp(-2t_{C_0}^2k_1k_2)$, uniformly in $(S_0,w)$, hence also unconditionally.

**(3)** By (1), $2\exp(-2t_{C_0}^2k_1k_2) = 2e^{-k_1\ln\frac{eN}{k_1}}e^{-k_2\ln\frac{eN}{k_2}}2^{-(\sigma+2)}\frac{\gamma_0}{4N^2}$. There are $|\mathcal{Z}|\binom{N}{k_1}\binom{N}{k_2}$ tuples with the given $(k_1,k_2)$; $\binom{N}{k} \le (eN/k)^k$ and $|\mathcal{Z}| < 2^{\sigma+2}$ cancel the three exponential factors, leaving at most $\gamma_0/(2N^2)$ per pair and $\gamma_0/2$ in total.

**(4)** $t_{C_0}^2 \le \frac{L}{2k_2} + \frac{L}{2k_1} + \frac{C_0}{2k_1k_2}$ since $k_i \ge 1$, and $1/k_i \le 2m_i$ by Lemma 5 bounds the three terms by $Lm_2$, $Lm_1$, $2C_0m_1m_2$.

**(5)** By concavity of $\sqrt{\cdot}$ and Lemma 4(ii),(iii), $\mathbb{E}[t_{C_0}] \le \sqrt{L(\mathbb{E}m_1 + \mathbb{E}m_2) + 2C_0\mathbb{E}[m_1m_2]} \le \sqrt{2\delta(L+C_0)}$. Numerically $L + C_0 = 1 + \ln N + (\sigma+2)\ln 2 + \ln 4 + 2\ln N + \ln\gamma_0^{-1} \le 3\ln N + 0.694\sigma + 3.78 + \ln\gamma_0^{-1}$; as $\sigma' \ge 2.885\ln N$ and $\sigma' \ge 2$ for $N \ge 2$, this is at most $3.63\sigma' + \ln\gamma_0^{-1}$, whence the claim. □

**Remark 4** (**$M$ occurs nowhere in $t_{C_0}$**). The alphabet enters only through $p_\theta$, which cancels, and through the i.i.d. structure of $\{f(\mathbf{u})\}$, which is alphabet-blind. There are $2^M$ possible tests, but once the observer is fixed and the revealed values conditioned away, only $|\mathcal{Z}| < 2^{\sigma+2}$ of them are reached.

**Lemma 8 (universal test class).** *Let $N,M \ge 2$, $\gamma_0 \in (0,1)$, and put $C_1 := M\ln 2 + \ln\frac{4N^2}{\gamma_0}$. With probability at least $1-\gamma_0$ over $f \leftarrow_\$$ Fun, simultaneously for all $k_1,k_2 \in [N]$, all $B_i$ with $|B_i| = k_i$, and* all *$\theta : [M] \to \{0,1\}$, writing $R := B_1 \times B_2$,*

$$\left|\frac{1}{k_1k_2}\sum_{\mathbf{u}\in R}(\theta(f(\mathbf{u})) - p_\theta)\right| \le t_{C_1}(k_1,k_2).$$

*Proof.* **(1)** Fix $k_1,k_2$, sets $B_i$ with $|B_i| = k_i$, and a function $\theta$ — all fixed before $f$ is drawn. For $\mathbf{u} \in R$ put $Y_\mathbf{u} := \theta(f(\mathbf{u})) - p_\theta$. The coordinates of $f$ are i.i.d. uniform on $[M]$, so the $k_1k_2$ variables $Y_\mathbf{u}$ are independent; each has mean zero exactly, since $\Pr\big[\theta(f(\mathbf{u})) = 1\big] = |\theta^{-1}(1)|/M = p_\theta$; and each lies in $\{-p_\theta, 1-p_\theta\}$, an interval of length 1. Assumption 2 with $\lambda := k_1k_2t_{C_1}$ gives failure probability at most $2\exp(-2t_{C_1}^2k_1k_2)$.

**(2)** By (1), $2t_{C_1}^2k_1k_2 = k_1\ln\frac{eN}{k_1} + k_2\ln\frac{eN}{k_2} + C_1$, so that bound equals $2(\frac{eN}{k_1})^{-k_1}(\frac{eN}{k_2})^{-k_2}2^{-M}\frac{\gamma_0}{4N^2}$.

**(3)** For the fixed pair $(k_1,k_2)$ the number of tuples $(B_1,B_2,\theta)$ is $\binom{N}{k_1}\binom{N}{k_2}2^M \le (\frac{eN}{k_1})^{k_1}(\frac{eN}{k_2})^{k_2}2^M$, by Assumption 2 and the fact that there are exactly $2^M$ functions $[M] \to \{0,1\}$. The three exponential factors cancel exactly, leaving at most $\gamma_0/(2N^2)$ per pair and $\gamma_0/2 \le \gamma_0$ over the at most $N^2$ pairs. □

**Remark 5 (the technique is classical; the two-source instance is not).** Lemma 8 is the two-source analogue of a textbook argument. Vadhan, *Pseudorandomness* (Foundations and Trends in TCS 7, 2012), Proposition 6.12, proves that a random function extracts from a fixed flat source by exactly this route: a union bound over *all* $2^M$ subsets $T \subseteq [M]$, a Chernoff bound for each fixed $T$, and the observation that the good event is a property of the function alone. Its Theorem 6.14 supplies the $\binom{N}{K} \le (Ne/K)^K$ counting that produces $t_c$; and the remark following Proposition 6.12 — that the number of flat $k$-sources is $\binom{N}{K} \approx N^K$, a larger double exponential — is the same one-source-fails observation as Remark 7 below. (Both verified against the monograph.)

What is *not* classical, and is where the novelty of this section sits, is Lemma 4 — that the posterior of $\mathbf{x}$ factorises because the sources are split, which is what makes a rectangle available at all — together with Lemma 6(a), whose Cauchy–Schwarz across the two coordinates yields $\delta\sqrt{M}$ rather than $\sqrt{M\delta}$ and has no one-source counterpart. Neither is available with a single source.

**Remark 6 (the quantifier order is the mechanism).** The good event is a property of $f$ alone, and on it the bound holds for every $\theta$ fixed in advance; substituting the particular $\theta_{\zeta,f}$, which does depend on $f$, is legitimate because a statement "for all $\theta$, $\Phi(f,\theta) \le t$" may be instantiated at any $\theta$ whatsoever. This is why no conditioning on inspected values is needed, why the sum runs over all of $R$, and why no revealing rule, budget, excised set or term $\Pi(\mathcal{S})$ appears. The trade against Lemma 7 is $\sigma + 2 \rightsquigarrow M$ inside a budget paid at rate $\delta\sqrt{2C_1}$, i.e. $\delta\sqrt{\sigma} \rightsquigarrow \delta\sqrt{M}$ in the final bound, and *not* $\sqrt{\sigma\delta} \rightsquigarrow \sqrt{M\delta}$.

**Remark 7 (where one source would break both).** With a single source emitting a point of $[N]^2$, Lemma 4(i) is unavailable and Lemma 5 flattens onto an arbitrary subset $B \subseteq [N]^2$ of size $k = \lfloor 1/m \rfloor$, not onto a rectangle. Step (3) would then range over $\binom{N^2}{k} \le (eN^2/k)^k$ sets, forcing $t^2 \ge \frac{1}{2}\ln(eN^2/k) \ge \frac{1}{2}$ since $k \le N^2$, so $t \ge 1/\sqrt{2}$ and the conclusion is vacuous, independently of $\delta$, $M$, $\sigma$. Lemma 6(a) collapses likewise, its Cauchy–Schwarz step being applied across the two coordinates. The failure occurs before any bound on the probability that a fixed set contains $\mathbf{x}$ is invoked, which is what the conjecture's own single-source counterexample requires.

## 5 The observer as a revealing rule

**Definition 4 (challenge resolution).** $D$ has *challenge resolution $M'$* if there is $h : [M] \to [M']$ such that for *every coin string $\rho$ of $D$*, every $f$, every $\zeta$, and all $v, v'$ with $h(v) = h(v')$, the runs $D^f_\rho(v,\zeta)$ and $D^f_\rho(v',\zeta)$ issue the same sequence of query *positions*; the output bit is unconstrained. Every $D$ has resolution $M$; resolution 1 means that, for each fixed coin string, the query positions ignore the challenge value. We may take $h$ surjective without loss.

The coin quantifier is part of the definition and not a technicality. The weaker condition that the query positions be challenge-independent *in law* defines a strictly larger class: at $N = M = 2$, $q = 1$, the observer that flips $\rho$ and queries $(1,1)$ when

$v \oplus \rho = 1$ and $(2,2)$ otherwise has query position uniform on $\{(1,1),(2,2)\}$ for both challenge values, yet each of its two coin fixings has resolution 2.

**Lemma 9.** *Let $D$ be deterministic, $q$-query, challenge-oblivious, of resolution $M'$. Then the procedures "$\mathcal{A}_\zeta$: for $j = 1, \ldots, M'$ run $D^f(v_j, \zeta)$ for a fixed $v_j \in h^{-1}(j)$, inspecting each queried cell not already known" form a revealing rule of budget $s = \min(qM', N^2)$, with $\theta_{\zeta,f}(v) := D^f(v,\zeta)$.*

*Proof.* $\mathcal{A}_\zeta$ inspects one coordinate at a time, the next a function of $\zeta$ and the values already seen: the loop and the representatives are fixed in advance, and inside run $j$ the next position is a function of $(v_j, \zeta)$ and previous answers. Each of the $M'$ runs issues at most $q$ positions, so at most $qM'$ cells, and at most $N^2$ exist. Finally, for arbitrary $v$ with $j = h(v)$, the run $D^f(v,\zeta)$ issues exactly the positions run $j$ issued, all inspected and recorded, so its whole execution — hence its output bit — is reconstructible from $(v, \zeta, \mathcal{S}, f|_{\mathcal{S}})$. Therefore the entire function $\theta_{\zeta,f}$, and $p_{\theta_{\zeta,f}}$, is determined by $(\zeta, \mathcal{S}, f|_{\mathcal{S}})$. □

**Lemma 10 (mass of the revealed set).** *Let $\mathcal{S}(f,\zeta) \subseteq [N]^2$ be a function of $(f,\zeta)$ with $|\mathcal{S}| \le s$. Then for $s \ge 1$*

$$\mathbb{E}\big[\min(\Pi_{f,\zeta}(\mathcal{S}(f,\zeta)),1)\big] \le \mu'(s) := \min\big(s\delta,\ 2(s\delta^2)^{1/3},\ 1\big),$$

*and the left side is $0$ for $s = 0$.*

*Proof.* By Lemma 4(i), $\Pi(\mathcal{S}) = \sum_{\mathbf{u}\in\mathcal{S}} \pi^1(u_1)\pi^2(u_2) \le s\, m_1 m_2$; expectations and Lemma 4(iii) give the arm $s\delta$, and the arm 1 is trivial. For the third, fix $\vartheta \in (0,1]$, bound by $s\vartheta$ on $\{m_1 m_2 \le \vartheta\}$ and by 1 off it, and use Lemma 6(b): $\mathbb{E}[\min(\Pi(\mathcal{S}),1)] \le g(\vartheta) := s\vartheta + \delta\vartheta^{-1/2}$. Then $g'(\vartheta) = s - \frac{1}{2}\delta\vartheta^{-3/2}$ vanishes at $\vartheta^* = (\delta/2s)^{2/3}$, which lies in $(0,1]$ since $\delta \le 1 \le 2s$, and $g'' > 0$; and $g(\vartheta^*) = (2^{-2/3} + 2^{1/3})(s\delta^2)^{1/3} \le 1.89(s\delta^2)^{1/3} \le 2(s\delta^2)^{1/3}$. □

The arm $2(s\delta^2)^{1/3}$ cannot be improved to $s\delta^2$, since $\mathbb{E}[m_1 m_2] \le \delta^2$ is false (Remark 3). The exponent $1/3$ is exactly what the $\sqrt{\vartheta}$ of Lemma 6(b) produces, and the bound is tight at $s = 1$ on that counterexample.

# 6 Bounds on the extraction advantage

**Lemma 11.** *For deterministic $D$ with associated $\theta_{\zeta,f}$, $\Pr[\mathsf{Real} = 1] = \mathbb{E}_{(f,\zeta)}\big[\mathbb{E}_{\mathbf{x}\sim\Pi_{f,\zeta}}[\theta_{\zeta,f}(f(\mathbf{x}))]\big]$ and $\Pr[\mathsf{Real}_0 = 1] = \mathbb{E}_{(f,\zeta)}[p_{\theta_{\zeta,f}}]$.*

*Proof.* Condition on $(H,\mathbf{z}) = (f,\zeta)$. In both experiments $D$'s oracle is $f$, so its output on challenge $v$ is $\theta_{\zeta,f}(v)$. In Real the challenge is $f(\mathbf{x})$ with $\mathbf{x} \sim \Pi_{f,\zeta}$ by Lemma 4; in $\mathsf{Real}_0$ it is uniform and independent, with $\mathbb{E}_{v \leftarrow_\$ [M]}[\theta(v)] = p_\theta$. □

**Proposition 1.** *For every $\gamma_0 \in (0,1)$ and deterministic $q$-query challenge-oblivious $D$ of resolution $M'$,*

$$\big|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\big| \le \gamma_0 + \sqrt{8\delta(\sigma' + \log\gamma_0^{-1})} + \mu'\big(\min(qM', N^2)\big).$$

*Proof.* By Lemma 11 and the triangle inequality the left side is at most $\mathbb{E}[\Delta]$ with $\Delta(f,\zeta) := |\mathbb{E}_\Pi[\theta(f(\mathbf{x})) - p_\theta]|$. Put $G_{f,\zeta}(\mathbf{u}) := (\theta(f(\mathbf{u})) - p_\theta)\mathbb{1}[\mathbf{u} \notin \mathcal{S}(f,\zeta)] \in [-1,1]$, so that $\Delta \le |\mathbb{E}_\Pi[G_{f,\zeta}]| + \Pi(\mathcal{S})$. Lemma 5 bounds $|\mathbb{E}_\Pi[G]|$ by the largest $|\mathbb{E}_{\mathrm{Unif}(B_1)\otimes\mathrm{Unif}(B_2)}[G]|$ over $|B_i| = k_i = \lfloor 1/m_i \rfloor$, and for such a rectangle that quantity equals $|\frac{1}{k_1k_2}\sum_{\mathbf{u}\in R\setminus\mathcal{S}}(\theta(f(\mathbf{u})) - p_\theta)|$, which on the good event of Lemma 7 is at most $t_{C_0}(k_1,k_2)$. As $\Delta \le 1$ always, $\mathbb{E}[\Delta] \le \Pr[\mathcal{E}^c] + \mathbb{E}[t_{C_0}] + \mathbb{E}[\min(\Pi(\mathcal{S}),1)]$, and the three terms are bounded by $\gamma_0$, by Lemma 7, and by $\mu'(s)$ via Lemmas 9 and 10. □

**Theorem 1 ($M$-free, resolution-restricted).** $\kappa^{\mathrm{na}}(q) \le 5\sqrt{\sigma'\delta} + q\delta$, *and* $\kappa(0) \le 5\sqrt{\sigma'\delta}$.

*Proof.* Both suprema are over observers permitted to randomise, whereas Proposition 1 is stated for deterministic $D$; by Lemma 3 the suprema are attained on deterministic observers, and $D'$ inherits $D$'s resolution, so restricting to deterministic resolution-1 observers loses nothing. If $\delta = 1$ then $\sqrt{\sigma'\delta} \ge \sqrt{2} > 1$ and there is nothing to prove; else take $\gamma_0 := \delta$. Since $\delta \ge 1/N$ we get $\log\gamma_0^{-1} \le \log N \le \sigma'/2$, so the middle term is at most $\sqrt{12\sigma'\delta} \le 3.47\sqrt{\sigma'\delta}$, and $\gamma_0 = \delta \le \sqrt{\sigma'\delta}$ as $\delta \le 1 \le \sigma'$. Resolution 1 gives $s \le q$ and $\mu'(q) \le q\delta$. For $q = 0$, $s = 0$ and every observer has resolution 1 vacuously. □

**Theorem 2 ($q$-free, every observer).** *Let $N,M \ge 2$ and $\gamma_0 \in (0,1)$. For* every *challenge-oblivious observer $D$ — of any query count, bounded or unbounded, and any challenge resolution —*

$$\big|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\big| \le \gamma_0 + \sqrt{2\delta\ln(eN)} + \delta\sqrt{2M\ln 2 + 2\ln\tfrac{4N^2}{\gamma_0}}.$$

*Proof.* By Lemma 3 we may assume $D$ deterministic; Lemma 3 imposes no bound on the query count and nothing below refers to one. With $\theta_{\zeta,f}$ and $p_\theta$ as in Lemma 11, set $\Delta(f,\zeta) := |\mathbb{E}_{\mathbf{u}\sim\Pi_{f,\zeta}}[\theta_{\zeta,f}(f(\mathbf{u})) - p_{\theta_{\zeta,f}}]|$; by Lemma 11 and the triangle inequality the left side is at most $\mathbb{E}[\Delta]$.

Fix $(f,\zeta)$ in the support and put $G_{f,\zeta}(\mathbf{u}) := \theta_{\zeta,f}(f(\mathbf{u})) - p_{\theta_{\zeta,f}} \in [-1,1]$. By Lemma 4 the law $\Pi_{f,\zeta}$ is a product, so Lemma 5 applies with this $G$ and gives

$$\Delta(f,\zeta) \le \max_{|B_i|=k_i}\Big|\frac{1}{k_1k_2}\sum_{\mathbf{u}\in B_1\times B_2}\big(\theta_{\zeta,f}(f(\mathbf{u})) - p_{\theta_{\zeta,f}}\big)\Big|.$$

Let $\mathcal{E}$ be the good event of Lemma 8, a property of $f$ alone, of probability at least $1-\gamma_0$. On $\mathcal{E}$ the maximum is at most $t_{C_1}(k_1,k_2)$: the pair $(k_1,k_2)$ lies in $[N]\times[N]$ by Lemma 5, the maximising $B_i$ are among the sets quantified over, and $\theta_{\zeta,f}$ is one particular element of $\{0,1\}^{[M]}$, over all of which Lemma 8 quantifies — this is Remark 6. Since $G \in [-1,1]$ gives $\Delta \le 1$ pointwise, and $t_{C_1} \ge 0$,

$$\mathbb{E}[\Delta] \le \Pr[\mathcal{E}^c] + \mathbb{E}[\mathbb{1}_{\mathcal{E}}t_{C_1}] \le \gamma_0 + \mathbb{E}[t_{C_1}] \le \gamma_0 + \sqrt{2L\delta} + \delta\sqrt{2C_1},$$

by Lemma 6(d) at $c = C_1$. □

For $M = 1$ the set $\mathsf{Fun}$ is a singleton and $\mathsf{Real}, \mathsf{Real}_0$ are the same experiment, so the advantage is 0 and the excluded case needs no bound.

**Corollary 1.** *For every $q \in \mathbb{N} \cup \{0\}$, $\kappa(q) \leq 5\sqrt{\sigma'\delta} + 2\delta\sqrt{M}$, with no restriction on the observer.*

*Proof.* If $\delta = 1$ then $5\sqrt{\sigma'\delta} \geq 5\sqrt{2} > 1 \geq \kappa(q)$; assume $\delta < 1$ and take $\gamma_0 := \delta$. Recall $\delta \geq 1/N$, $N \geq 2$, and $\sigma' \geq 2\log N \geq 2$, so $\ln N \leq \frac{\ln 2}{2}\sigma' \leq 0.3466\sigma'$ and $1 \leq \sigma'/2$. Then $2\delta L = 2\delta(1 + \ln N) \leq 1.6932\sigma'\delta$, so $\sqrt{2\delta L} \leq 1.3012\sqrt{\sigma'\delta}$; and, using $\ln(1/\delta) \leq \ln N$,

$$2C_1 = 2M\ln 2 + 2\ln 4 + 4\ln N + 2\ln\tfrac{1}{\delta} \leq 1.3863M + 2.7726 + 6\ln N \leq 1.3863M + 3.4657\sigma',$$

whence $\delta\sqrt{2C_1} \leq 1.1774\delta\sqrt{M} + 1.8617\delta\sqrt{\sigma'} \leq 1.1774\delta\sqrt{M} + 1.8617\sqrt{\sigma'\delta}$, using $\delta\sqrt{\sigma'} = \sqrt{\delta}\sqrt{\sigma'\delta} \leq \sqrt{\sigma'\delta}$. Finally $\gamma_0 = \delta \leq \sqrt{\sigma'\delta}$ as $\delta \leq 1 \leq \sigma'$. Adding, $(1 + 1.3012 + 1.8617)\sqrt{\sigma'\delta} + 1.1774\delta\sqrt{M} \leq 5\sqrt{\sigma'\delta} + 2\delta\sqrt{M}$. □

**Theorem 3 (the revealing-rule arm, collected).** *For every $\gamma_0$ and every deterministic $q$-query challenge-oblivious $D$ of resolution $M'$, Proposition 1 with Lemma 6(d) in place of Lemma 7's cruder expectation bound gives*

$$\kappa(q) \leq \sqrt{2\delta\ln(eN)} + 4\delta\sqrt{\sigma'} + \mu'\big(\min(qM, N^2)\big),$$

*and the same with $\mu'(q)$ at $M' = 1$, which bounds $\kappa^{\mathrm{na}}(q)$.*

*Proof.* Run Proposition 1 with $\mathbb{E}[t_{C_0}]$ bounded by Lemma 6(d) at $c = C_0$ rather than by concavity on the whole expression, and take $\gamma_0 = \delta$. Then $\ln(1/\delta) \leq \ln N$ and $2C_0 \leq 1.3863\sigma + 5.5452 + 6\ln N \leq 6.238\sigma'$, giving $\delta\sqrt{2C_0} \leq 2.498\delta\sqrt{\sigma'}$; and $\gamma_0 = \delta \leq 0.708\delta\sqrt{\sigma'}$, the two together at most $3.21\delta\sqrt{\sigma'} \leq 4\delta\sqrt{\sigma'}$. □

**Corollary 2 (the two arms).** $\kappa(q) \leq 5\sqrt{\sigma'\delta} + \min\{\mu'(\min(qM, N^2)),\ 2\delta\sqrt{M}\}$, *the first arm granting Theorem 3 and the second unconditional by Corollary 1. Both leading parts are at most $5\sqrt{\sigma'\delta}$: $4.164$ for the second, and $1.302\sqrt{\sigma'\delta} + 3.21\delta\sqrt{\sigma'} \leq 4.512\sqrt{\sigma'\delta}$ for the first, using $\delta\sqrt{\sigma'} \leq \sqrt{\sigma'\delta}$.*

**Remark 8 (which arm wins).** Suppose $q \geq 1$, $M \geq 4$, and $2\delta\sqrt{M} < 1$. Then the second arm is the smaller: against $\mu'$'s arm 1 immediately; against $2(s\delta^2)^{1/3}$ with $s \leq qM$ because $2\delta\sqrt{M} \leq 2(qM\delta^2)^{1/3}$ iff $\delta\sqrt{M} \leq q$, and $\delta\sqrt{M} < 1/2 < q$; against $s\delta \leq qM\delta$ because $q\sqrt{M} \geq 2$. The cap $s = N^2$ never binds usefully, since $\mu'(N^2) = 1$: both $N^2\delta \geq N \geq 2$ and $2(N\delta)^{2/3} \geq 2$, as $\delta \geq 1/N$. So for every $q \geq 1$ and $M \geq 4$ the second arm dominates wherever either says anything; the first survives only at $q = 0$, where $\mu'(0) = 0$ is exact and $\delta\sqrt{M}$ is pure loss, and in the corner $M \in \{2, 3\}$.

# 7 A lower bound

**Proposition 2.** *Let $N \geq 2$ and $2 \leq M \leq N^2/2$. Let $S_1, S_2$ each draw $x_i \leftarrow_\$ [N]$ on private coins and output $(x_i, \varepsilon)$. Then the pair is split, $\sigma_1 = \sigma_2 = 0$, $\sigma' = 2\log N$, and it is $\delta$-unpredictable with $\delta = 1/N$. There is a deterministic challenge-oblivious $D^*$, making $N^2$ queries and of challenge resolution $1$, with $|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]| \geq \delta\sqrt{M}/(4\sqrt{2})$.*

*Proof.* **(1)** The coins are independent, so the pair is split. Here $\mathbf{z} = (\varepsilon, \varepsilon)$ always and each $x_i$ is uniform on $[N]$ and independent of $H$ and of $x_{3-i}$. A predictor is an arbitrary function of $(\mathbf{z}, H)$, so its output is independent of $x_i$ and $\Pr\left[\mathsf{P}^H(\mathbf{z}) = x_i\right] = 1/N$ for every $\mathsf{P}$.

**(2)** $D^*$ queries all $N^2$ cells, so holds $f$; forms $\nu(v) := |f^{-1}(v)|/N^2$; and on challenge $v$ outputs $\mathbb{1}[\nu(v) > 1/M]$. Its query positions are the same for every challenge value, so its resolution is 1.

**(3)** Condition on $H = f$. In Real, $\mathbf{x}$ is uniform on $[N]^2$ and independent of $f$, so the challenge $f(\mathbf{x})$ has law exactly $\nu$; in $\mathsf{Real}_0$ it is uniform on $[M]$. With $A_f := \{v : \nu(v) > 1/M\}$,

$$\Pr\left[\mathsf{Real} = 1 \mid f\right] - \Pr\left[\mathsf{Real}_0 = 1 \mid f\right] = \sum_v \left(\nu(v) - \tfrac{1}{M}\right)^+ = \mathrm{SD}\left(\nu, \mathrm{Unif}([M])\right) \ \geq 0,$$

so no cancellation occurs on averaging and the advantage equals $\mathbb{E}_f[\mathrm{SD}(\nu, \mathrm{Unif}([M]))]$.

**(4)** *Claim: let $Y = \sum_{i=1}^n Y_i$ with $Y_i$ i.i.d., $\mathbb{E}Y_1 = 0$, $|Y_1| \leq 1$, and $s^2 := \mathbb{E}Y^2 = n\mathbb{E}Y_1^2$. If $s \geq 1$ then $\mathbb{E}|Y| \geq s/2$.* Discarding the terms with an index to an odd power, which vanish, $\mathbb{E}Y^4 = n\mathbb{E}Y_1^4 + 3n(n-1)(\mathbb{E}Y_1^2)^2 \leq n\mathbb{E}Y_1^2 + 3n^2(\mathbb{E}Y_1^2)^2 = s^2 + 3s^4$, using $\mathbb{E}Y_1^4 \leq \mathbb{E}Y_1^2$, valid as $|Y_1| \leq 1$. Hölder with exponents $3/2$ and $3$ gives $s^2 = \mathbb{E}[|Y|^{2/3}|Y|^{4/3}] \leq (\mathbb{E}|Y|)^{2/3}(\mathbb{E}Y^4)^{1/3}$, so $\mathbb{E}|Y| \geq s^3/\sqrt{s^2 + 3s^4} \geq s^3/(2s^2) = s/2$ for $s \geq 1$.

**(5)** Put $n := N^2$, $p := 1/M$, $X_v := |f^{-1}(v)|$. Over uniform $f$ the summands $\mathbb{1}[f(\mathbf{u}) = v]$ are i.i.d. Bernoulli$(p)$, so $X_v \sim \mathrm{Bin}(n, p)$ and each centred summand lies in $[-p, 1-p]$, of absolute value at most 1. As $\mathrm{SD}(\nu, \mathrm{Unif}([M])) = \frac{1}{2n}\sum_v |X_v - np|$ and the $X_v$ are identically distributed, $\mathbb{E}_f[\mathrm{SD}] = \frac{M}{2n}\mathbb{E}|X - np|$. Here $s^2 = np(1-p) = \frac{N^2}{M}(1 - \frac{1}{M}) \geq \frac{N^2}{2M}$ for $M \geq 2$, and $s^2 \geq 1$ since $M \leq N^2/2$, so (4) gives $\mathbb{E}|X - np| \geq \frac{N}{2\sqrt{2M}}$ and $\mathbb{E}_f[\mathrm{SD}] \geq \frac{M}{2N^2} \cdot \frac{N}{2\sqrt{2M}} = \frac{\sqrt{M}}{4\sqrt{2}N}$. □

So on the diagonal $\delta = 1/N$, Corollary 1's $M$-dependence is pinned to within $8\sqrt{2} < 12$. The family fixes $\delta = 1/N$ and $2 \leq M \leq N^2/2$, and licenses nothing off that diagonal — nor anything at $q < N^2$, which matters in §12.

# 8 Presampling, in oracle-indexed consistent form

**Lemma 12.** *Let $N, M \geq 2$, $P \in \mathbb{N}$, $\gamma \in (0,1)$. There is a family $\mathcal{Y}^{P,\gamma} = \{Y_{f,\zeta}\}$ indexed by $f \in \mathsf{Fun}$ and $\zeta \in (\{0,1\}^*)^2$ — with $Y_{f,\zeta}$ uniform on $\mathsf{Fun}$ at every $\zeta$ outside the support of $\mathbf{z}$, so the family is total, no experiment reaching such a $\zeta$ — depending only on $(S_1, S_2, P, \gamma)$, in which every $Y_{f,\zeta}$ is a $P$-mixture consistent with $f$ — indeed a single $P$-bit-fixing source — such that for every $q$ and every $q$-query observer taking $\mathbf{z}$ and an independent uniform value of $[M]$,*

$$\left|\Pr[\mathsf{Real}_0 = 1] - \Pr[\mathsf{Dec}_0 = 1]\right| \leq \frac{q(\sigma + 2 + 2\log\gamma^{-1})}{P} + 2\gamma.$$

*Proof.* **Step 1.** For $\zeta$ with $\Pr[\mathbf{z} = \zeta] > 0$ let $\mu_\zeta$ be the law of $H$ given $\mathbf{z} = \zeta$ and $S_\zeta := N^2 \log M - H_\infty(\mu_\zeta)$. Since $\mu_\zeta(f) = \Pr\left[H = f\right]\Pr\left[\mathbf{z} = \zeta \mid H = f\right]/\Pr[\mathbf{z} = \zeta] \leq$

$(|\mathsf{Fun}|\Pr[\mathbf{z}=\zeta])^{-1}$, we get $S_\zeta \le \log(1/\Pr[\mathbf{z}=\zeta])$. With $\bar{S} := \sigma + 2 + \log\gamma^{-1}$ and $\mathcal{B} := \{\zeta : S_\zeta > \bar{S}\}$, every $\zeta \in \mathcal{B}$ has $\Pr[\mathbf{z}=\zeta] < \gamma 2^{-(\sigma+2)}$, so $\Pr[\mathbf{z}\in\mathcal{B}] < |\mathcal{Z}|\gamma 2^{-(\sigma+2)} < \gamma$.

**Step 2.** Fix $\zeta$ and set $\delta_\zeta := (S_\zeta + \log\gamma^{-1})/(P\log M)$. If $\delta_\zeta > 1$ the asserted bound exceeds $q\log M \ge q$, hence is vacuous for $q \ge 1$, and for $q = 0$ both experiments give the observer a uniform challenge and no oracle access; set $Y_{f,\zeta}$ uniform. Otherwise apply Assumption 2 to $\mu_\zeta$ with parameter $\delta_\zeta$; since $(S_\zeta + \log\gamma^{-1})/(\delta_\zeta \log M) = P$ it yields $\mu_\zeta = \sum_j \lambda_j X_j + \lambda_{\text{fin}} Y_{\text{fin}}$ with $\lambda_{\text{fin}} \le \gamma$, each $X_j$ a $(P, 1-\delta_\zeta)$-dense source, all supports pairwise disjoint, and each $X_j$ fixing its set $I_j$, $|I_j| \le P$, to values every $f$ in its support takes. Let $Y_j$ be the corresponding $P$-bit-fixing source and define $Y_{f,\zeta} := Y_j$ for the unique $j$ with $f \in \operatorname{supp} X_j$, and $Y_{f,\zeta} :=$ uniform otherwise. Consistency is immediate; the construction reads only $(f,\zeta)$ and the decomposition, which depends only on $(S_1, S_2, P, \gamma)$. Write $J = J(f,\zeta)$.

**Step 3.** Fix $\zeta \notin \mathcal{B}$ with $\delta_\zeta \le 1$ and condition on $J = j$. Since $\lambda_j X_j$ is the restriction of $\mu_\zeta$ to $\operatorname{supp} X_j$, the conditional law of $H$ is exactly $X_j$; in $\mathsf{Dec}_0$ the oracle is drawn freshly from $Y_j$. The challenge is uniform and independent, so conditioning on it leaves a fixed $q$-query adaptive distinguisher between $X_j$ and $Y_j$; Assumption 2 bounds its advantage by $q\delta_\zeta \log M = q(S_\zeta + \log\gamma^{-1})/P$, and averaging over the challenge preserves this.

**Step 4.** Averaging over $j$ and bounding the residue event, of conditional probability at most $\gamma$, by 1, then averaging over $\zeta$ and bounding $\mathcal{B}$'s contribution by $\Pr[\mathbf{z}\in\mathcal{B}] < \gamma$ while using $S_\zeta \le \bar{S}$ off $\mathcal{B}$, gives the claim. □

For $M = 1$, $\mathsf{Fun}$ is a singleton, so $\mathsf{Real}_0$ and $\mathsf{Dec}_0$ coincide and the bound holds with left side 0; the construction divides by $\log M$ and is stated for $M \ge 2$ only.

# 9 From extraction to decomposition

**Theorem 4.** *With $\mathcal{Y} := \mathcal{Y}^{P,\gamma/2}$, for every $q$ and every $q$-query challenge-oblivious $D$, writing $\varepsilon(D) := |\Pr[\mathsf{Real}=1] - \Pr[\mathsf{Real}_0 = 1]|$ for that same $D$,*

$$\mathsf{Adv}_{\mathcal{Y},D} \le \varepsilon(D) + \frac{2(\sigma' + \log\gamma^{-1})q}{P} + \gamma + P\delta + q\delta.$$

*Proof.* Interpolate $\mathsf{G}_0 = \mathsf{Real}$, $\mathsf{G}_1 = \mathsf{Real}_0$, $\mathsf{G}_2 = \mathsf{Dec}_0$, $\mathsf{G}_3 = \mathsf{Dec}$.

$|\mathsf{G}_0 - \mathsf{G}_1| = \varepsilon(D)$ by definition, for the fixed $D$ the theorem is instantiated at; no supremum is taken here.

$|\mathsf{G}_1 - \mathsf{G}_2|$: Lemma 12 with slack $\gamma/2$ gives $q(\sigma + 2 + 2\log(2/\gamma))/P + \gamma$; as $N \ge 2$ gives $\sigma + 2 \le \sigma'$ and $\sigma' \ge 2$, the numerator is at most $2\sigma' + 2\log\gamma^{-1}$. For $q \ge P$ that term already exceeds $2\sigma' > 1$, so no hypothesis $q < P$ is needed.

$|\mathsf{G}_2 - \mathsf{G}_3| \le P\delta + q\delta$: draw $H$, $(\mathbf{x},\mathbf{z})$ and $J$ with fixed set $I_J$; let $H^\circ \sim Y_J$ and $U \leftarrow_\$ [M]$ be independent of everything else; let $H^*$ be $H^\circ$ with its value at $\mathbf{x}$ overwritten by $U$ when $\mathbf{x} \notin I_J$. As $J$ is a function of $(H,\mathbf{z})$, $H^\circ$ is independent of $\mathbf{x}$ given $(H,\mathbf{z},J)$; conditioning on $\mathbf{x} = \mathbf{u} \notin I_J$, the source $Y_J$ is uniform and independent at $\mathbf{u}$, so both $H^\circ$ and $H^*$ are distributed as $Y_J$. Running $\mathsf{G}_3$ with oracle $H^*$ and $\mathsf{G}_2$ with oracle $H^\circ$ and challenge $U$, the two executions receive the same input on $\{\mathbf{x} \notin I_J\}$, since there $H^*(\mathbf{x}) = U$, and their oracles agree off $\mathbf{x}$. So they diverge only if $\mathbf{x} \in I_J$

or $D$ queries $\mathbf{x}$; being identical up to the step before a first query at $\mathbf{x}$, that event has equal probability in both and may be measured in $\mathsf{G}_2$. Lemma 1 gives $\Pr[\mathbf{x} \in I_J] \le P\delta$ — its hypotheses hold, $\mathcal{Y}$ depending on $(S_1, S_2, P, \gamma)$ alone, being indexed by $(f, \zeta)$, and having its component drawn independently of $\mathbf{x}$ given $(H, \mathbf{z})$, since $J$ is a function of $(H, \mathbf{z})$. Lemma 2 gives $\Pr[\mathbf{x} \in Q] \le q\delta$ in $\mathsf{G}_2$, its predictor being simulable because the challenge there is uniform and independent. □

**Remark 9 (where the $P$-cap enters, and that it is one step).** $P\delta$ enters the entire chain at exactly the step above, via Lemma 1. It grows in $P$ while the target's first term $\sim \sigma' q^+/P$ shrinks, and balancing them at $P \approx \sqrt{\sigma' q^+/\delta}$ gives $\sqrt{\sigma' q^+ \delta}$, which is precisely the target's second term. That is why (H1) below caps $P$ at the balance point, and it is the only reason.

Note also what the step pays for: on $\{\mathbf{x} \in I_J\}$ consistency forces $H^*(\mathbf{x}) = H(\mathbf{x})$, the challenge Real supplies — so the route pays $P\delta$ for the event on which Dec's challenge is *correct*. Consistency is used nowhere in this section. That is the sense in which the cap is a defect of the route rather than of the statement, and it is why §11 does not attack $P\delta$ at all.

# 10 Reducing the hypotheses

**Lemma 13.** *Let $N \ge 2$. If $8\sqrt{\sigma' q^+ \delta} \ge 1$ then any conclusion of the form $\mathsf{Adv} \le 2Bq^+/P + 8\sqrt{\sigma' q^+ \delta} + \gamma$ holds trivially, its right-hand side being at least $1 \ge \mathsf{Adv}$. Otherwise $q^+\delta \le \sigma'$ holds automatically.*

*Proof.* $\mathsf{Adv}$ is a difference of two probabilities, so $\mathsf{Adv} \le 1$; and the right-hand side is at least $8\sqrt{\sigma' q^+ \delta}$, every summand being non-negative. That disposes of the first case. In the second, $8\sqrt{\sigma' q^+ \delta} < 1$ gives $\sigma' q^+ \delta < 1/64$, hence $q^+\delta < 1/(64\sigma')$. Since $\sigma' \ge 2$, $q^+\delta < 1/128 < 2 \le \sigma'$. □

**Theorem 5.** *Let $N, M \ge 2$, $\gamma \in (0,1)$, $P \in \mathbb{N}$, $q \in \mathbb{N} \cup \{0\}$, and let $\mathcal{Y} := \mathcal{Y}^{P,\gamma/2}$ be the family of Lemma 12 — built from $(S_1, S_2, P, \gamma)$ alone, before $q$, indexed by $(f, \zeta)$, each member a $P$-mixture consistent with its index. Suppose*

(H1) $P \le \sqrt{\sigma' q^+/\delta} + 1$, *and*

(H2) $\min\{\mu'(\min(qM, N^2)),\ 2\delta\sqrt{M}\} \le \sqrt{\sigma' q^+ \delta}$.

*Then for every $q$-query challenge-oblivious observer $D$, with no restriction on challenge resolution,*

$$\mathsf{Adv}_{\mathcal{Y},D} \le \frac{2(\sigma' + \log \gamma^{-1}) q^+}{P} + 8\sqrt{\sigma' q^+ \delta} + \gamma.$$

*Proof.* By Lemma 13 we may assume $8\sqrt{\sigma' q^+ \delta} < 1$, the other case being trivial. That gives both $\sigma' q^+ \delta < 1/64$, used in the sharpened sub-bounds below, and $q^+\delta \le \sigma'$. Theorem 4 gives, for the same fixed $D$,

$$\mathsf{Adv}_{\mathcal{Y},D} \le \varepsilon(D) + \frac{2(\sigma' + \log \gamma^{-1}) q}{P} + \gamma + P\delta + q\delta,$$

and the second term is at most the target's first, as $q \le q^+$. Four terms remain.

*The extraction term.* Corollary 2 applies to $D$ with no hypothesis and gives $\varepsilon(D) \le \kappa(q) \le 5\sqrt{\sigma'\delta} + \min\{\mu'(\min(qM, N^2)), 2\delta\sqrt{M}\}$. Here $5\sqrt{\sigma'\delta} \le 5S$, and the minimum is at most $S$ by (H2), so $\varepsilon(D) \le 6S$.

*Two sharpened sub-bounds.* Since $\sigma' q^+\delta < 1/64$ we have $q^+\delta < 1/(64\sigma')$, so $q^+\delta \le \frac{1}{8\sigma'}S$ and, using $\sigma' \ge 2$,

$$q\delta \le q^+\delta \le \tfrac{1}{16}S.$$

Likewise $\delta < 1/(64\sigma' q^+)$ gives $\delta \le \frac{1}{8\sigma' q^+}S \le \frac{1}{16}S$. Both are worth having: the crude bounds $q\delta \le S$ and $\delta \le S$ each spend a whole unit of the target's 8, and there is no unit to spare.

*The fixed-set term.* With $t_q = \sqrt{\sigma' q^+/\delta}$, so that $t_q\delta = S$, (H1) gives $P \le t_q + 1$ and hence $P\delta \le t_q\delta + \delta \le S + \frac{1}{16}S = \frac{17}{16}S$.

*The query term.* $q\delta \le \frac{1}{16}S$, above.

Adding, $6 + \frac{17}{16} + \frac{1}{16} = 7\frac{1}{8} \le 8$. □

**Corollary 3 (at $q = 0$ there is no restriction on $M$).** *Let $N, M \ge 2$, $\gamma \in (0,1)$ and $P \le 3\sqrt{\sigma'/\delta}$. Then for every 0-query challenge-oblivious observer and every $M$,* $\mathsf{Adv}_{\mathcal{Y},D} \le 2(\sigma' + \log\gamma^{-1})/P + 8\sqrt{\sigma'\delta} + \gamma$.

*Proof.* (H2) at $q = 0$ reads $\min\{\mu'(\min(0, N^2)), 2\delta\sqrt{M}\} = \min\{\mu'(0), 2\delta\sqrt{M}\}$, and $\mu'(0) = 0$, so (H2) holds for every $M$. In Theorem 4 at $q = 0$ the chain is $\mathsf{Adv} \le \varepsilon(D) + \gamma + P\delta$, the two $q$-carrying terms vanishing. Theorem 1 gives $\kappa(0) \le 5\sqrt{\sigma'\delta}$ with no hypothesis on $M$ and none on resolution — at $q = 0$, $s = 0$ and every observer has resolution 1 vacuously — so $\varepsilon(D) \le 5\sqrt{\sigma'\delta}$, leaving three units of the target's $8\sqrt{\sigma'\delta}$ for $P\delta$, and $P\delta \le 3\sqrt{\sigma'\delta}$ iff $P \le 3\sqrt{\sigma'/\delta}$. □

**Remark 10 (the region for Conjecture 1 is the intersection over $q$).** Theorem 5 is a statement *per* $q$: it fixes $P$ and $q$ jointly and then quantifies over $q$-query observers. Conjecture 1 is not. It fixes one family per $(P, \gamma)$ — before $q$ — and demands the bound at *every* $q$. So the conjecture holds at $(P, \gamma)$ exactly when (H1) and (H2) hold at every $q$, and that is the intersection, not the region. Both are weakest at small $q$. (H1)'s right side increases in $q$, so it binds at $q = 0$: $P \le \sqrt{\sigma'/\delta} + 1$. (H2) is vacuous at $q = 0$ and, via its $2\delta\sqrt{M}$ arm, reads $M \le \sigma' q^+/(4\delta)$ for $q \ge 1$, binding at $q = 1$: $M \le \sigma'/(2\delta)$. This distinction is the single most-mistaken point in the lineage and is used below.

## 11 The capping construction

This is the new content, and it does not attack $P\delta$. It observes that one never has to pay it at the requested $P$.

Given a requested $P \in \mathbb{N}$ and $\gamma \in (0,1)$, put

$$\gamma_0 := \max(\gamma, N^{-2}), \qquad P_0 := \min\big(P, \lceil t_q \rceil\big), \qquad \mathcal{Y} := \mathcal{Y}^{P_0, \gamma_0/2}.$$

**Lemma H0 (admissibility).** Every member of $\mathcal{Y}$ is a $P$-mixture consistent with its index; $\mathcal{Y}$ is indexed by $(f,\zeta)$ and by nothing else; and Theorem 5 applies to it at $(P_0,\gamma_0)$.

*Proof.* Each member is a $P_0$-mixture, so each component has fixed set $I$ with $|I| \le P_0 \le P$. Definition 2 requires $|I| \le P$ and nothing more, so the same witness $(I,a)$ that proves $P_0$-bit-fixing proves $P$-bit-fixing. The uniformity and independence conditions off $I$ refer to the same unchanged $I$, and consistency is a property of the fixed values $a$, so both survive verbatim. The index set is that of $\mathcal{Y}^{P_0,\gamma_0/2}$, namely $\mathsf{Fun} \times (\{0,1\}^*)^2$, as Remark 1 requires. For Theorem 5 at $(P_0,\gamma_0)$: $\gamma_0 \in (0,1)$ since $\gamma < 1$ and $N^{-2} \le 1/4$; (H1) reads $P_0 \le t_q + 1$ and holds because $P_0 \le \lceil t_q \rceil \le t_q + 1$; and (H2) does not mention $P$, so it is unaffected. Finally $P_0 \ge 1$: $\sigma' \ge 2$, $q^+ \ge 1$ and $\delta \le 1$ give $t_q \ge \sqrt{2} > 1$, so $\lceil t_q \rceil \ge 2$, and $P \ge 1$. □

**Lemma H1 (three slack bounds).** $\log \gamma_0^{-1} \le \log \gamma^{-1}$; $\log \gamma_0^{-1} \le \sigma'$, hence $\sigma' + \log \gamma_0^{-1} \le 2\sigma'$; and $N^{-2} \le \frac{1}{4} S$.

*Proof.* $\gamma_0 \ge \gamma$ gives the first, and it is what keeps $c$ at 2: without it the clamp would inflate the leading term rather than the additive one. $\gamma_0 \ge N^{-2}$ gives $\log \gamma_0^{-1} \le 2 \log N \le \sigma'$. For the third, $\delta \ge 1/N$, $\sigma' \ge 2$ — which needs $N \ge 2$ — and $q^+ \ge 1$ give $S \ge \sqrt{2/N}$, whence

$$N^{-2} \le \tfrac{1}{4} S \iff N^{-2} \le \tfrac{1}{4}\sqrt{2/N} \iff N^{-3} \le \tfrac{1}{2},$$

true for every $N \ge 2$. □

**Theorem H (at fixed $q$, no hypothesis on $P$).** Let $N, M \ge 2$, fix $q \in \mathbb{N} \cup \{0\}$ and suppose (H2). For every $P \in \mathbb{N}$ and every $\gamma \in (0,1)$, the family $\mathcal{Y}$ above consists of $P$-mixtures consistent with their indices, depends only on $(S_1, S_2, P, \gamma, q)$, and satisfies, for every $q$-query challenge-oblivious observer $D$ and with no restriction on challenge resolution,

$$\mathsf{Adv}_{\mathcal{Y},D} \;\le\; \frac{2(\sigma' + \log \gamma^{-1})\, q^+}{P} \;+\; 13\, S \;+\; \gamma.$$

*Proof.* Lemma H0 licenses Theorem 5 at $(P_0,\gamma_0)$, giving $\mathsf{Adv} \le 2(\sigma' + \log \gamma_0^{-1}) q^+ / P_0 + 8S + \gamma_0$. Two cases.

*Case (a), $P \le \lceil t_q \rceil$.* Then $P_0 = P$, and by Lemma H1 $\log \gamma_0^{-1} \le \log \gamma^{-1}$, so the first term is at most $2Bq^+/P$, which is the target's own first term with $c = 2$. For the additive term, $\gamma_0 \le \gamma + N^{-2} \le \gamma + \frac{1}{4} S$. Total: $2Bq^+/P + 8\frac{1}{4} S + \gamma$.

*Case (b), $P > \lceil t_q \rceil$.* Then $P_0 = \lceil t_q \rceil \ge t_q$, and by Lemma H1 $\sigma' + \log \gamma_0^{-1} \le 2\sigma'$, so

$$\frac{2(\sigma' + \log \gamma_0^{-1}) q^+}{P_0} \;\le\; \frac{4\sigma' q^+}{t_q} \;=\; 4\sigma' q^+ \sqrt{\frac{\delta}{\sigma' q^+}} \;=\; 4S.$$

With $\gamma_0 \le \gamma + \frac{1}{4} S$ again, the total is at most $12\frac{1}{4} S + \gamma \le 13S + \gamma$, and the target's first term is non-negative.

Both cases are dominated by $2Bq^+/P + 13S + \gamma$. □

**Remark 11 (what this is, and is not).** This is Conjecture 1's *display* at $c = 2$, $C = 13$, with no hypothesis on $P$. It is *not* the conjecture, which requires the family to depend only on $(S_1, S_2, P, \gamma)$; a $q$-dependent family satisfies a relaxation of it, and Remark 2 is explicit that the order is part of the statement. Only one thing changes against the conjecture: the family may depend on $q$, through $P_0 = \min(P, \lceil t_q \rceil)$ and nothing else. It still may not depend on $D$, on $\mathbf{x}$, or on the challenge; it is still one family per $(P, \gamma, q)$, still indexed by $(f, \zeta)$, still consistent. Against Theorem 5 the trade is exact: (H1) is deleted outright and $C$ moves $8 \to 13$, while $c$ does not move. Of the five extra units, four are case (b)'s capping and a quarter is the $\gamma$-clamp; three quarters go unused.

**Theorem H′ (one $q$-free family, every $P$, at a cost of $\sqrt{q^+}$).** Let $N, M \geq 2$ and suppose (H2) holds at *every* $q \in \mathbb{N} \cup \{0\}$ — equivalently, by Remark 10, $M \leq \sigma'/(2\delta)$, the intersection binding at $q = 1$. Then the family $\mathcal{Y}' := \mathcal{Y}^{P_0', \gamma_0/2}$ with $P_0' := \min(P, \lceil t_0 \rceil)$ depends only on $(S_1, S_2, P, \gamma)$ — not on $q$ — consists of $P$-mixtures consistent with their indices, and satisfies, for every $P \in \mathbb{N}$, every $q \in \mathbb{N} \cup \{0\}$ and every $q$-query challenge-oblivious observer $D$,

$$\mathsf{Adv}_{\mathcal{Y}', D} \;\leq\; \frac{2(\sigma' + \log \gamma^{-1})\, q^+}{P} \;+\; \big(4\sqrt{q^+} + 9\big) S \;+\; \gamma.$$

*Proof.* As in Lemma H0, with (H1) at $P_0'$ holding for *every* $q$ at once, since $P_0' \leq \lceil t_0 \rceil \leq t_0 + 1 \leq t_q + 1$ by $t_0 \leq t_q$. (H2) is *not* uniform in $q$ and is supplied at every $q$ by hypothesis. Theorem 5 then applies at $(P_0', \gamma_0)$ for each $q$. Case (a), $P \leq \lceil t_0 \rceil$, is as before. Case (b), $P > \lceil t_0 \rceil$: now $P_0' \geq t_0 = \sqrt{\sigma'/\delta}$, so

$$\frac{4\sigma' q^+}{t_0} = 4\sigma' q^+ \sqrt{\frac{\delta}{\sigma'}} = 4 q^+ \sqrt{\sigma' \delta} = 4\sqrt{q^+} \cdot S,$$

using $S = \sqrt{q^+}\sqrt{\sigma'\delta}$. Adding $8S$ and $\gamma_0 \leq \gamma + \frac{1}{4} S$ gives the claim, with a quarter-unit to spare. ◻

**Remark 12 (the hypothesis is strictly stronger than (H2)).** An earlier draft wrote "suppose (H2) holds at $q$" and then quantified the conclusion over *every* $q$, leaving $q$ free in the hypothesis and rebinding it in the conclusion. Three readings, all defective. Read at one $q$, the hypothesis is *vacuously* satisfiable at $q = 0$, since $\mu'(0) = 0$ makes (H2) free there, so the theorem would deliver the conjecture for every $M$ and contradict §13. Read per-$q$, it is true but is then not Conjecture 1, which demands one family good at every $q$. Only the intersection reading is both true and sufficient, and it was not written down.

The distinction is not idle. A non-vacuous witness: $N = 2^{12}$, $\sigma = 0$ (so $\sigma' = 24$), $\delta = 2^{-12}$, $M = 2^{16}$. Then $8\sqrt{\sigma' q^+ \delta} < 1$ at both $q = 0$ and $q = 1$; (H2) holds at $q = 0$; and at $q = 1$ it fails on both arms, $2\delta\sqrt{M} = 0.125$ and $\mu'(\min(M, N^2)) = 0.315$, against $S = 0.108$.

**Remark 13 (relation to the converse construction, and what is not shown).** The Contract's own converse remark gives a $q$-free family with $\mathsf{Adv} \leq \kappa(q) + Aq/P_0 + \gamma + P_0\delta + q\delta$ at $P_0 = \min(P, \lceil \sqrt{A/\delta} \rceil)$, $A := \sigma + 2 + \log(1/\gamma)$, concluding the conjecture "for

$P \le \sqrt{A/\delta}$, and for every $P$ when $q = O(1)$", the two directions separated by $\Theta(\sqrt{q^+})$ for growing $q$. Two things must be said carefully.

*It is not the same cap.* $A$ carries a $\log(1/\gamma)$ that $\sigma' = \sigma + 2\log N$ does not, and has $+2$ where $\sigma'$ has $2\log N$. The two coincide only at $\gamma = N^{-2}$ and diverge without bound as $\gamma \downarrow 0$. Theorem H′ is an *analogue* of that construction, run against Theorem 5 rather than against $\kappa$ directly. It is not a reproduction of it.

*Only one direction is shown.* It is tempting to conclude that the $\Theta(\sqrt{q^+})$ separation *is* the price of $q$-independence, since capping at $t_q$ instead of $t_0$ removes it. That claims a necessity nothing here establishes: no lower bound against $q$-free families appears anywhere, and §13 lists finding such a family — or showing none exists — as open. Moreover the two quantities are different in kind. The converse remark's $\sqrt{q^+}$ is a separation *in the value of $P$* between two directions of a biconditional; the $\sqrt{q^+}$ in Theorem H′ is a *degradation of the constant $C$ at fixed $P$*. They share an exponent, not a mechanism.

## 12 What (H2) is, and whether applications satisfy it

(H2) is not a condition about decomposition. It is inherited from routing the decomposition through extraction, and it enters at exactly one step: the extraction term of Theorem 5, where Corollary 2's residue must fit inside one unit of the budget $S$.

The second arm has a clean reading. By Proposition 2's calculation, a random function's image distribution $\nu(v) = |f^{-1}(v)|/N^2$ is lumpy: throwing the pair's $\approx 1/\delta^2$ effective probability mass into $M$ buckets gives a balls-in-bins deviation $\approx \sqrt{M\delta^2} = \delta\sqrt{M}$. The real challenge is drawn from that lumpy law and the decomposed one is uniform off the fixed set, so the lumpiness genuinely appears in Adv. Exploiting it, however, requires knowing *which* values are over-represented, and that means learning $f$. That is why $q = 0$ is free for every $M$ (Corollary 3), and why $D^*$ in Proposition 2 spends $N^2$ queries.

When $D$ has full challenge resolution — as a general distinguisher $D^H(y, \mathbf{z})$ does, since it may steer its queries on $y$ — the second arm is the smaller for $q \ge 1$, $M \ge 4$ by Remark 8, and (H2) becomes, in bits,

$$\log M \;\le\; k + \log(\sigma' q^+) - 2, \qquad k := \log(1/\delta).$$

*Do not ask for more output bits than one source's unpredictability*, plus a logarithmic allowance. The $q$-dependence is only additive-logarithmic: $q$ from $2^0$ to $2^{128}$ moves the allowance by 127 bits. So the corner where (H2) binds is not really about query budgets — it is about asking for a long output.

The two failing rows differ in kind and only one matters. Row 4 fails (H2), but the bound there is $2^{-27}$ anyway, so nothing is lost that one had. Rows 5 and 6 are the real gap: security is still $2^{-91}$ and (H2) blocks the statement purely for asking for a long output. That is precisely the sponge and Merkle–Damgård streaming use which motivated wanting a decomposition route rather than a compression argument in the first place.

| setting | $k$ | output | allowed | (H2) | bound |
|---|---|---|---|---|---|
| 128-bit key, 256-bit-unpredictable sources | 256 | 128 | $\leq 328$ | yes | $2^{-91}$ |
| 256-bit key, 256-bit-unpredictable sources | 256 | 256 | $\leq 328$ | yes | $2^{-91}$ |
| 128-bit key, 128-bit-unpredictable sources | 128 | 128 | $\leq 200$ | yes | $2^{-27}$ |
| 256-bit key, 128-bit-unpredictable sources | 128 | 256 | $\leq 200$ | **no** | $2^{-27}$ |
| 512-bit output, 256-bit sources | 256 | 512 | $\leq 328$ | **no** | $2^{-91}$ |
| 1 Mbit stream, 256-bit sources | 256 | $2^{20}$ | $\leq 328$ | **no** | $2^{-91}$ |

Table 1: (H2) at application parameters, with $\sigma = 64$, $q = 2^{64}$, $\log N = 512$. Computed by `c/0010/campaign/checks/h2-applications.py`.

**Remark 14 (the evidence for** (H2) **does not reach the corner that binds).** Proposition 2, the only lower bound here, exhibits an observer making $N^2$ queries. But $\kappa$ is monotone in $q$ — a $q$-query observer may ignore queries — so a lower bound at $q = N^2$ constrains $\kappa(1)$ not at all. Meanwhile (H2) is free at $q = N^2$, permitting $M \leq \sigma' N^2/(4\delta)$, and binds at small $q$. *So nothing here gives any evidence that $\kappa(q)$ carries $M$-dependence where* (H2) *actually bites.* The sharp question is therefore narrower than "can the first arm of Corollary 2 be made $M$-free": does $\kappa(q)$ have *any* $M$-dependence for small $q$? Exhibit a $q$-query lower bound carrying $M$ for $q \ll N^2$, or prove $\kappa(q) \leq O(\sqrt{\sigma'\delta}) + (M\text{-free})$ there.

## 13 What is settled, and what is not

**Settled, on the region** (H2). The $P$-axis, at fixed $q$, in full: no hypothesis on $P$ survives in Theorem H. And at $q = 0$, (H2) is vacuous for every $M$ since $\mu'(0) = 0$, so Theorem H is unconditional there.

**Not settled.**

- *Conjecture 1 as stated.* Theorem H's family depends on $q$; Theorem H′'s constant does. Neither is the conjecture, which wants both a $q$-free family and an absolute $C$. The honest residue on the $P$-axis is a factor $\sqrt{q^+}$ in the constant under $q$-free families, on $M \leq \sigma'/(2\delta)$.
- *Whether the $\sqrt{q^+}$ is necessary.* Only achievability is shown; see Remark 13.
- *The $M$-corner.* (H2) is carried, not discharged, except at $q = 0$. Remark 14 sharpens the question.
- *Whether $\delta\sqrt{M}$ is tight off the diagonal.* Proposition 2 licenses nothing off $\delta = 1/N$, and nothing at $q < N^2$.

**A redirection**. By Remark 9, $P\delta$ enters at one step, and the natural programme is to remove it so that a *large* fixed set can be afforded. At fixed $q$ that work is unnecessary: one never needs a fixed set larger than $\lceil t_q \rceil$, so $P\delta$ never needs to exceed $S + \delta$. Such a programme is aimed solely at the $q$-free statement, where the cap must sit at $t_0$ and the $\sqrt{q^+}$ is real.

# 14 Verification status

Verification is not uniform across this document, and reading it as though it were would be the main way to be misled by it.

**§11 has had six independent blind referee passes** in fresh context, spanning two model families: five on the arm itself (lenses: the definitional step; arithmetic and constants; quantifier order; case coverage and degenerate instances; citation fidelity) and one on three proposed simplifications. All five returned DEFECTS. Lemma H0 was attacked by four referees independently, each trying to construct a reading of Definition 2 on which it fails; none could. The arithmetic referee re-derived every inequality by hand, reproduced the machine check's grid sizes exactly including collision counts, found *no computational error*, and confirmed that $c = 2$ and $C = 13$ follow from the displayed inequalities rather than from the grid. Seven class-(A) and class-(C) findings were upheld and repaired, every one a claim *about* the results rather than a result: the malformed hypothesis of Theorem H′ (Remark 12); an identification of the $\sqrt{q^+}$ with the converse remark's separation (Remark 13); the unstated standing hypothesis $N, M \geq 2$; a machine check described as reaching $q \leq 10^7$ whose vacuity filter admitted nothing above $q^+ = 410$; and summary lines scoped to a stronger statement than the body proves. The revision carrying those repairs has itself had *no* pass.

**§10 has had two passes**, both DEFECTS, both repaired; the current form has had none.

**§§6–7 have had none**. This is the weakest link and it is load-bearing. Corollary 2 supplies the extraction term of Theorem 5, on which everything in §11 rests, and it sits in an arm with zero verification passes. Worse, the one referee objection ever formed about its constant was *overruled* rather than upheld, and never re-tested. A reader who wants one thing checked should check Corollary 2.

**§§3–5 and §8 have had one pass**, verdict CLEAN, with six class-(C) findings that were never adjudicated.

**The external inputs have had none**. Assumptions 2–2 are quoted from source cards transcribed from local copies, and no pass has ever checked a card against the paper it summarises. Three consequences of Assumption 2 used in §8 are not stated in the published claim in that form.

**Lemmas 1 and 2 are unproved here and unrefereed anywhere**. They are the Contract's, they are load-bearing inside Theorem 4 at exactly the step where the $P$-cap enters, and no verification pass has examined them, the Contract having been handed to every referee as the yardstick rather than as an object of review.

**No prior-art search has ever been run on this campaign**, so the novelty of Theorem 2, Corollary 1 and Proposition 2 — all unconditional — is unchecked. And no counterexample search has been run against any current artifact.